Application E2 - Problem Statement:

A ball of mass $m = 500\ \mathrm{g}$ is thrown up an inclined plane of $\alpha = 0.1\ \mathrm{rad}$ with an initial speed $V_A = 3\ \mathrm{m/s}$. The ball reaches point B with a speed $V_B = 1\ \mathrm{m/s}$ after a time $\Delta t = 1.2\ \mathrm{s}$. The reference level for Gravitational Potential Energy (GPE) is at A. (We will use $g = 10\ \mathrm{m/s^2}$ as in your other examples).

a. Applying the general form of Newton's second law, determine the intensity of the frictional force (f). b. Determine the distance D (from A to B) traveled by the ball. c. Calculate the amount of heat produced. Convert the answer to calories.

Part a: Finding the Frictional Force (f)

We will use the **Theorem of Linear Momentum** (Impulse-Momentum) as requested, which relates force and time.

1. Define System, States, and Units:

- System: The ball.
- Mass (m): $500\ \mathrm{g} = 0.5\ \mathrm{kg}$
- Time Interval (Δt): $1.2\ \mathrm{s}$
- Coordinate System: Let's define the positive x-axis ($\vec{i}$) as pointing up the incline, from A to B.
- Initial State (at A): $\vec{V}_A = 3\vec{i}\ \mathrm{m/s} \implies \vec{P}_A = m\vec{V}_A = (0.5)(3\vec{i}) = 1.5\vec{i}\ \mathrm{kg \cdot m/s}$
- Final State (at B): $\vec{V}_B = 1\vec{i}\ \mathrm{m/s} \implies \vec{P}_B = m\vec{V}_B = (0.5)(1\vec{i}) = 0.5\vec{i}\ \mathrm{kg \cdot m/s}$

2. Analyze the External Forces ($\Sigma\vec{F}_{ext}$): We only need to consider the forces parallel to the incline (along our x-axis).

- Gravitational Force (Weight) Component ($\vec{W}_x$): Acts down the incline. Its magnitude is $mg\sin(\alpha)$, so $\vec{W}_x = -mg\sin(\alpha)\vec{i}$.
- Frictional Force ($\vec{f}$): Opposes the motion (which is up the incline), so it also acts down the incline. $\vec{f} = -f\vec{i}$.
- Net Force (parallel to incline): $\Sigma\vec{F}_{ext,x} = (-mg\sin(\alpha) - f)\vec{i}$

3. Apply the Linear Momentum Theorem:

$$\Delta\vec{P} = \Sigma\vec{F}_{ext} \cdot \Delta t$$

We'll look at the x-components:

$$\vec{P}_B - \vec{P}_A = (\Sigma\vec{F}_{ext,x}) \cdot \Delta t$$

$$(0.5\vec{i}) - (1.5\vec{i}) = (-mg\sin(\alpha) - f)\vec{i} \cdot (1.2)$$

Now, looking at the scalar components:

$$-1.0 = (-mg\sin(\alpha) - f) \cdot 1.2$$

4. Solve for f:

$$-1.0 = (-(0.5)(10)\sin(0.1) - f) \cdot 1.2$$

Since $\alpha = 0.1 \text{ rad}$ *is a small angle,* $\sin(0.1) \approx 0.1$ *(or* 0.0998 *to be more precise). We'll use* 0.0998*.*

$$-1.0 = (-5 \cdot 0.0998 - f) \cdot 1.2$$

$$-1.0 = (-0.499 - f) \cdot 1.2$$

$$\frac{-1.0}{1.2} = -0.499 - f$$

$$-0.833 = -0.499 - f$$

$$f = 0.833 - 0.499$$

$$f \approx 0.334 \text{ N}$$

Part b: Finding the Distance (D)

We will use the **Work-Energy Theorem** (non-conservation of ΔME) since it relates force and distance.

$$\Delta ME = W_{nc} \implies ME_B - ME_A = W_f$$

1. Define Energy States:

- **Non-Conservative Work (W_{nc}):** Work done by friction $\vec{f}$. Since friction opposes the displacement $\vec{D}$, $W_f = -f \cdot D$.
- State A (Initial):
  - $KE_A = \frac{1}{2}mV_A^2 = \frac{1}{2}(0.5)(3)^2 = 2.25 \text{ J}$
  - $GPE_A = 0$ (at reference level).
  - $ME_A = 2.25 \text{ J}$
- State B (Final):
  - $KE_B = \frac{1}{2}mV_B^2 = \frac{1}{2}(0.5)(1)^2 = 0.25 \text{ J}$
  - $GPE_B = mgh_B$. The height h_B is related to D by $h_B = D\sin(\alpha)$.
  - $GPE_B = mgD\sin(\alpha) = (0.5)(10)D\sin(0.1) \approx 0.499D$.
  - $ME_B = 0.25 + 0.499D$

2. Apply the Theorem and Solve for D:

$$ME_B - ME_A = W_f$$

$$(0.25 + 0.499D) - (2.25) = -f \cdot D$$

$$0.25 + 0.499D - 2.25 = -0.334D$$

$$-2.0 + 0.499D = -0.334D$$

$$-2.0 = -0.334D - 0.499D$$

$$-2.0 = -0.833D$$

$$D = \frac{-2.0}{-0.833}$$

$$D \approx 2.4 \text{ m}$$

Part c: Finding the Heat Produced (Q)

The heat produced (Q) by friction is the energy dissipated, which is equal to the negative of the work done by friction (W_f).

1. Calculate Heat in Joules (J):

$$Q = -W_f$$

$$Q = -(-f \cdot D) = f \cdot D$$

$$Q = (0.334 \text{ N}) \cdot (2.4 \text{ m})$$

$$Q \approx 0.802 \text{ J}$$

2. Convert to Calories (cal): Given the conversion $1 \text{ cal} \approx 4.18 \text{ J}$.

$$Q \text{ (cal)} = \frac{Q \text{ (J)}}{4.18 \text{ J/cal}}$$

$$Q \text{ (cal)} = \frac{0.802}{4.18}$$